Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis

Network for Sustainable Food Transitions (NEST)

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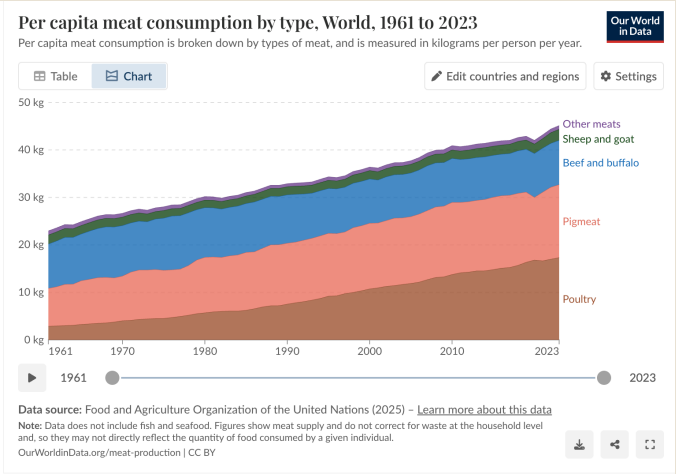
Fact 1: meat consumption is rising across every category



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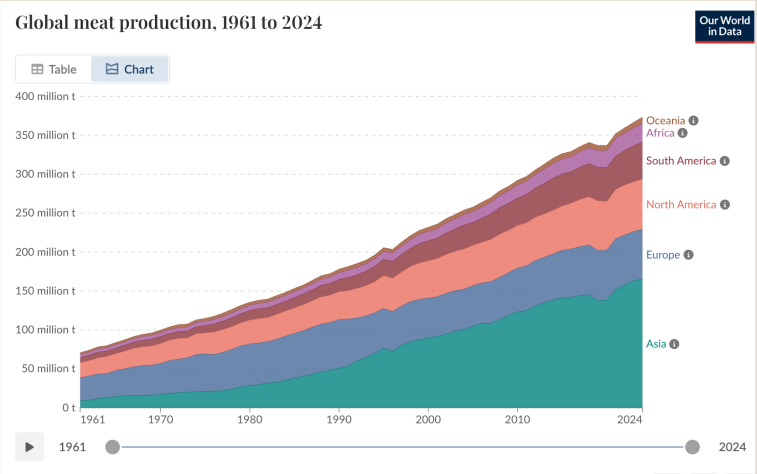
. . . everywhere



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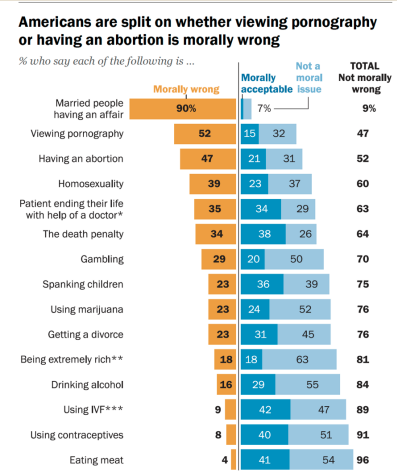
Fact 2: people think eating meat is fine



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Fact 3: transformational effects don't get discovered in the social sciences because they're already obvious

Daniel Lakens, talking about a putative effect size of d = 2 (the effect of hunger on judge's behavior):

> *"If hunger had an effect on our mental resources of this magnitude, our society would fall into minor chaos every day at 11:45... Just like manufacturers take size differences between men and women into account when producing items such as golf clubs or watches, we would stop teaching in the time before lunch, doctors would not schedule surgery, and driving before lunch would be illegal.* ***If a psychological effect is this big, we don't need to discover it and publish it in a scientific journal — you would already know it exists.****"*

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So how about this meta-analysis

- I worked on a meta-analytic review of interventions to get people to eat less MAP.
- We only looked at rigorous studies (RCTs with > 50 subjects that measured MAP consumption directly at least a day after treatment began)
- As of 12/2023, 35 papers met our criteria

Full Length Article

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What interventions have been tried?

- Almost everything tries to:
  - Tell people meat is bad (environmental/health/animal welfare)
  - Make plant-based options more salient (eye-level on menus)
  - Make animal-based options less convenient
  - Tell people plant-based options are tasty and popular
  - Exploit psychological tendencies (social norms, commitment)
- Overall, small & sparse effects

Table 1. Main findings.

Approach	N (Studies)	N (Estimates)	SMD	τ	95% CIs	p val
Overall	41	112	0.07	0.08	[0.02, 0.12]	0.007
Theory						
Choice Architecture	2	3	0.21	0	[-0.99, 1.42]	0.267
Psychology	9	12	0.11	0.03	[-0.03, 0.24]	0.092
Persuasion	25	77	0.07	0.09	[0.01, 0.13]	0.023
Persuasion & Psychology	11	20	0.11	0.19	[-0.06, 0.28]	0.189
Types of Persuasion						
Animal Welfare	16	65	0.03	0.04	[-0.02, 0.07]	0.189
Environment	16	29	0.09	0.12	[-0.02, 0.19]	0.095
Health	18	29	0.08	0.09	[-0.01, 0.17]	0.064

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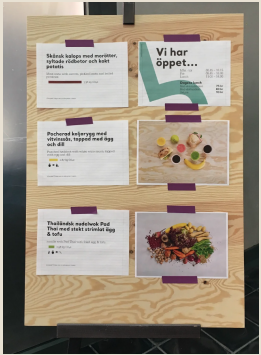
Choice architecture in particular: very few credible designs

- Only 2 studies, 3 estimates in our sample
- Most studies lack real-world follow-up: we don't know if cafeteria-based nudges change what people actually buy over time
- Lab and online studies dominate; field evidence is thin
- Pooled SMD = 0.21, but CI includes 0 ($[-0.99, 1.42]$) because there are, again, two papers



example menu from Andersson, O., & Nelander, L. (2021). Nudge the lunch: A field experiment testing menu-primacy effects on lunch choices. *Games*, 12(1), 2.

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Hi Isabel :)

- Time to stir the pot!
- What you call field experiments, I would call field *studies*. An experiment needs random assignment.
- Looking through your citations, *are* there any experiments that measure outcomes with any delay at all? I think Ginn & Sparkman (2024) is the only one that even theoretically would qualify and I don't see that in the citations
- was also surprised not to see Coker et al. (2022), "A dynamic social norm messaging intervention to reduce meat consumption: A randomized cross-over trial in retail store restaurants"
- But most importantly, it's really hard to gauge how much we should buy the causal assumptions in non-randomized before-after papers
  - Often it's clear that a lot of things changed, e.g. the number of meals served changes dramatically between periods
- Question for Isabel/anyone: *should* we grant these identification strategies the benefit of the doubt? I say no, the default is to assume bias. What do you all

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Conclusion: meaningfully reducing consumption of meat and animal products is an unsolved problem

- A lot of advocates want a villain — the big bad meat corporations, captured regulators, whatever — someone to defeat.
- I don't think that's the right model. There's no Lex Luthor here.
- People just really like eating meat, and they genuinely don't think it's morally problematic. (See: slide 3.)
- Our challenge is to persuade them otherwise — and that's going to be a lot of work.
- That's OK. That's the work we signed up for.

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